DOCKING-BASED PRIORITISATION OF RICEBERRY BRAN CONSTITUENTS AGAINST HUMAN α-GLUCOSIDASE AND α-AMYLASE FOR BY-PRODUCT VALORISATION

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A BY-PRODUCT WORTH MINING

Starch becomes glucose via pancreatic **α-amylase** and intestinal **α-glucosidase**. Acarbose slows both, but is poorly tolerated (flatulence 74% vs 29% on placebo, US label).

Riceberry bran is a low-value by-product of Thai rice milling. Pigmented-rice anthocyanins are already known **from in vitro work** to inhibit both enzymes (Akkarachiyasit et al., 2010).

Bran carries many candidates at once, and a food ingredient must also **dissolve where the enzymes are**. These have never been compared on the human enzymes.

THE QUESTION

Riceberry bran, a low-value by-product of Thai rice mills, carries anthocyanins, γ-oryzanol, vitamin E and phenolic acids.

Which fraction should a food project extract and assay first? That is the question this screen answers.

11 bran constituents ranked on both human enzymes, scoring predicted binding and solubility together

MATERIALS AND METHODS

1 Targets
Human MGAM (PDB 2QMJ, 1.9 Å) and pancreatic α-amylase (PDB 3BAJ, 2.1 Å), both crystallised with acarbose

2 Protein prep
PDBFixer: waters and ions removed, missing atoms and hydrogens added at pH 7.4; Meeko PDBQT

3 Ligands
12 structures from PubChem; RDKit ETKDG + MMFF94 optimisation; Meeko

4 Docking
AutoDock Vina 1.2.7 in the acarbose site, exhaustiveness 16, 3 independent seeds (78 runs)

5 Analysis
ΔΔG vs acarbose; contacts ≤ 4 Å and H-bonds ≤ 3.5 Å per residue; ligand efficiency; score–size regression; logP

Protocol validation: acarbose re-docked from a randomised start returned to its MGAM crystal pose in every run (RMSD 2.04–2.06 Å, n = 3).

RESULTS · RANKING THE BRAN

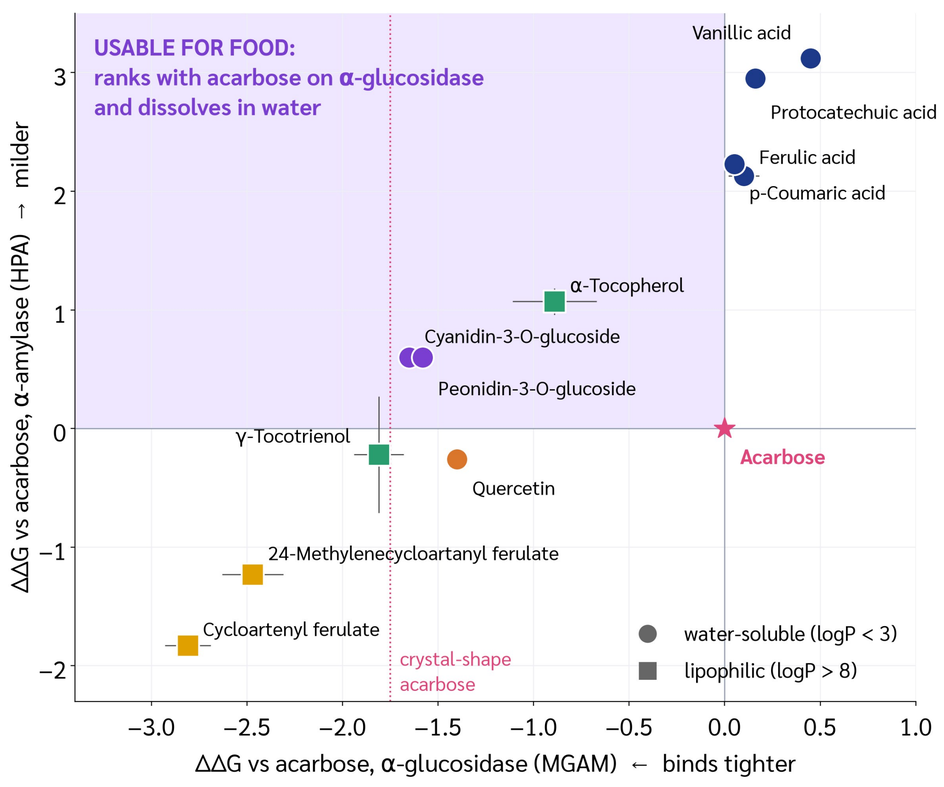


Fig. 1 Each constituent placed against acarbose on both enzymes (ΔΔG, mean of 3 seeds). Circles dissolve in water, squares do not. The shaded area marks where a usable food candidate would sit.

Only the anthocyanins fall in the shaded region: as good as acarbose on α-glucosidase, milder on α-amylase, and soluble in water.

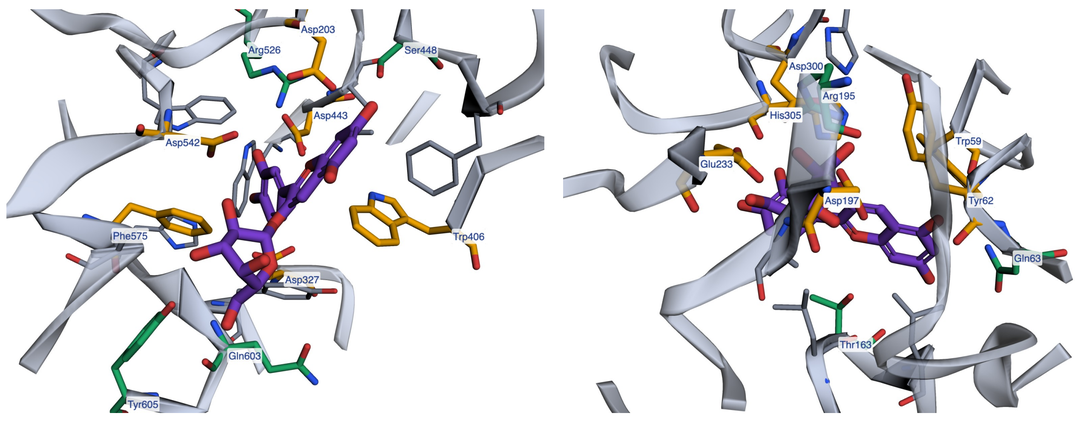


Fig. 2 Cyanidin-3-O-glucoside (purple) in α-glucosidase (left) and α-amylase (right). Gold = catalytic residues, green = hydrogen-bond partners.

The tightest binder is not the usable one

γ–Oryzanol esters topped the ranking (cycloartenyl ferulate –8.87 / –10.52 kcal/mol), but with logP > 10 they will not dissolve in the gut lumen, and they sit in the bran's oil fraction.

Anthocyanins are the only soluble front-runners

Cyanidin–3–O–glucoside (–7.71 / –8.09) and peonidin–3–O–glucoside (–7.64 / –8.09) ranked next, hydrogen-bonded the catalytic acids Asp443/Asp542 and Asp197/Glu233, and were the only high-ranking compounds with logP < 0.7.

Size inflates the score, so efficiency was scored too

Across the panel the predicted affinity rose with molecular size (Pearson r = 0.73, p = 0.007 on MGAM; r = 0.88, p = 0.0002 on HPA). Per heavy atom the anthocyanins bind more efficiently than the γ-oryzanol esters (0.24 vs 0.20 kcal/mol/atom on MGAM) and than acarbose (0.14).

FROM SCREEN TO PRODUCT

① Take this fraction

The **acidified aqueous extract** of fresh Riceberry bran — the anthocyanin-bearing fraction.

Not whole bran, not the oil fraction: the γ-oryzanol that ranks highest sits in the oil and will not dissolve.

Bran must be fresh; it turns rancid within weeks.

② Prove it on the bench

Two-enzyme IC_{50}, not one. Run α-glucosidase and α-amylase assays side by side against acarbose.

Report a **selectivity index** (IC_{50} amylase ÷ IC_{50} glucosidase) so potency and selectivity are measured, not inferred.

Quantify anthocyanin by pH-differential (520/700 nm) as cyanidin-3-glucoside equivalents.

③ Build it into food

Anthocyanins survive **acid and short heat**, not long baking. The prototype is a no-added-sugar jelly drink: pH 3.2–3.5, 80 °C for 1 min, hot-filled.

Three extract levels (20/30/40%), 9-point hedonic test, n ≥ 30.

Label claims stay factual: no health claim without FDA approval.

CONCLUSION

Scoring binding **together with solubility** changes which molecule a food project should chase. The highest-scoring constituents of Riceberry bran, the **γ-oryzanol esters**, cannot dissolve where the enzymes work; the **anthocyanins** rank just below them and are the only front-runners that can.

This turns a rice-mill by-product into one defined fraction to extract and one assay that would settle it, instead of a vague claim about rice bran being healthy.

What this cannot tell you. Docking ranks predicted binding, not inhibitory potency or selectivity; differences below ~1 kcal/mol sit inside the scoring error. Prior in vitro work established that these anthocyanins do inhibit both enzymes, and the two-enzyme IC_{50} comparison above is what would rank them. No health claim is made here.

Rotate the poses in 3D

Structures PDB 2QMJ (Sim et al., J Mol Biol 2008) and 3BAJ (Maurus et al., Biochemistry 2008). **Software** AutoDock Vina 1.2.7 (Eberhardt et al., JCIM 2021); Meeko; RDKit; PDBFixer. Ligands from PubChem. **Prior in vitro data** Akkarachiyasit et al., Int J Mol Sci 2010;11:3387. **Acarbose tolerability** US prescribing information. All scripts and raw results via the QR link.

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